

Apex Cloud Services - Kubernetes Platform Capacity Plan

Q4 2026-Q1 2027

1. Executive Summary

This document outlines the capacity expansion plan for Apex Cloud Services Kubernetes platform supporting Q4 2026 and Q1 2027 business growth targets. Led by Senior SRE Corey Holloway.

2. Current State January 2026

Metric	Current	Q4 2026 Target	Q1 2027 Target
Worker Nodes	48	72	96
Total vCPUs	192	288	384
Memory (GB)	768	1152	1536
Storage (TB)	12	18	24
Availability SLO	99.95%	99.97%	99.97%
Pod Startup Latency (s)	4.2	3.5	3.0

3. Risk Areas

- Node pool autoscaler lag during spikes: 90s warmup causes P2 incidents.
- PVC provisioner slow path under concurrent deploys: root cause INC-2025-063.
- Etcd three-node quorum: five-node migration underway ahead of Q4 peak.
- Ingress pod OOM risk at peak: memory limits need revision per INC-2026-001.

4. Mitigations

- Pre-warm node pool to 115 percent of projected peak; evaluate KEDA.
- Explicit storageClass in all deployment manifests per runbook post-INC-2025-063.
- Etcd five-node migration production rollout targeting Q4 2026 completion.
- Update HPA min replica floor for all ingress deployments.

5. Approval

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